

Ans1.

➤ Integer(int) :-

- Integer is used to store whole numbers.
- It does not contain a decimal point.
- Integer can be negative, positive or zero.
- It is used for mathematical calculations.

Example:-

Num = 32

➤ Float(float) :-

- Float is used to store numbers with decimal points.
- Float can also be negative or positive.

Example:-

Pi =3.14

➤ String(str):-

- String is used to store text, words, sentences and numbers.
- Anything written inside double quotes(" ") and triple quotes("""" """).

Example:-

Name = "Divya"

➤ Boolean(bool):-

- Boolean is used to represent True or False values.
- It is mainly used in decision- making and conditions.

Example:-

a=5

b=6

print(a>b)

output: false

Ans4.

Five commonly used string methods :-

i)upper() :

converts all letters of a string to uppercase.

Example:

address = "Pilani"

print(address.upper())

ii)lower():

converts all letters of a string to lowercase.

Example:

name = "DIVYA"

```
print(name.lower())
```

iii)replace():

It replaces one word or character with another.

Example:

```
intro = "I love java"
```

```
print(intro.replace("java","Python"))
```

iv)strip():

It removes extra spaces from the beginning and end of string.

Example:

```
text = " hello "
```

```
print(text.strip())
```

v)startswith():

check whether a string starts with a specific word or character.

It returns True or False .

Example:

```
Temp = "python"
```

```
print(Temp.startswith("py"))
```

Ans7.

Artificial Intelligence (AI) is a technology that allows computers and machines to think ,learn, and make decisions like humans.

✓ Importance of AI:-

- It saves time and effort.
- It helps people do work faster.
- It reduces mistakes.
- It makes daily tasks easier.
- It increases productivity.

Four real life applications:-

- ❖ Voice Assistants – Siri and Google Assistant answer questions and follow voice commands.
- ❖ Healthcare – AI helps doctors detect diseases.
- ❖ Self-Driving Cars – AI helps cars drive safely without a human driver.
- ❖ Online Recommendations – AI suggests movies, songs, and videos based on our interests.

Ans8.

1. ChatGPT – AI:

ChatGPT is an AI because it can understand questions and give answers like a human.

2. Google Maps Route Prediction – AI:

Google Maps uses AI to analyze traffic conditions and suggest the fastest route.

3. Calculator – Not AI:

A calculator only performs calculations according to fixed instructions. It cannot learn or make decisions.

4. Netflix Recommendations – AI:

Netflix uses AI to study your viewing history and recommend movies or shows you may like.

5. Voice Assistants (Alexa/Siri) – AI:

Alexa and Siri use AI to understand voice commands and respond to users.